PG No. 39 $S_4(c)$ $\bar{4}$ [tetragonal] (internal polar dipole)

* Harmonics for rank 0

$\vec{\mathbb{Q}}_0^{(1,1)}[q](A)$

** symmetry

1

** expression

$$\frac{\sqrt{3}Q_x x}{3}+\frac{\sqrt{3}Q_y y}{3}+\frac{\sqrt{3}Q_z z}{3}$$

* Harmonics for rank 1

$\vec{\mathbb{G}}_1^{(1,0)}[q](A)$

** symmetry

z

** expression

$$\frac{\sqrt{2}Q_x y}{2}-\frac{\sqrt{2}Q_y x}{2}$$

$\vec{\mathbb{Q}}_1^{(1,-1)}[q](B)$

** symmetry

z

** expression

Q_z

$\vec{\mathbb{Q}}_1^{(1,1)}[q](B)$

** symmetry

z

** expression

$$\frac{3\sqrt{10}Q_x xz}{10}+\frac{3\sqrt{10}Q_y yz}{10}-\frac{\sqrt{10}Q_z\left(x^2+y^2-2z^2\right)}{10}$$

$\vec{\mathbb{Q}}_1^{(1,-1)}[q](E^{(a)})$

** symmetry

$$\frac{\sqrt{2}x}{2}+\frac{\sqrt{2}iy}{2}$$

** expression

$$\frac{\sqrt{2}Q_x}{2}+\frac{\sqrt{2}iQ_y}{2}$$

$\vec{\mathbb{Q}}_1^{(1,1)}[q](E^{(a)})$

** symmetry

$$\frac{\sqrt{2}x}{2}+\frac{\sqrt{2}iy}{2}$$

** expression

$$\frac{\sqrt{5}Q_x\left(2x^2+3ixy-y^2-z^2\right)}{10}-\frac{\sqrt{5}iQ_y\left(x^2+3ixy-2y^2+z^2\right)}{10}+\frac{3\sqrt{5}Q_z z\left(x+iy\right)}{10}$$

$\vec{\mathbb{G}}_1^{(1,0)}[q](E^{(a)})$

** symmetry

$$\frac{\sqrt{2}x}{2}-\frac{\sqrt{2}iy}{2}$$

** expression

$$\frac{iQ_x z}{2}+\frac{Q_y z}{2}-\frac{iQ_z\left(x-iy\right)}{2}$$

$\vec{\mathbb{Q}}_1^{(1,-1)}[q](E^{(b)})$

** symmetry

$$\frac{\sqrt{2}x}{2}-\frac{\sqrt{2}iy}{2}$$

** expression

$$\frac{\sqrt{2}Q_x}{2} - \frac{\sqrt{2}iQ_y}{2}$$

$\vec{\mathbb{Q}}_1^{(1,1)}[q](E^{(b)})$
** symmetry

$$\frac{\sqrt{2}x}{2} - \frac{\sqrt{2}iy}{2}$$

** expression

$$\frac{\sqrt{5}Q_x\left(2x^2 - 3ixy - y^2 - z^2\right)}{10} + \frac{\sqrt{5}iQ_y\left(x^2 - 3ixy - 2y^2 + z^2\right)}{10} + \frac{3\sqrt{5}Q_z z\left(x - iy\right)}{10}$$

$\vec{\mathbb{G}}_1^{(1,0)}[q](E^{(b)})$
** symmetry

$$\frac{\sqrt{2}x}{2} + \frac{\sqrt{2}iy}{2}$$

** expression

$$-\frac{iQ_x z}{2} + \frac{Q_y z}{2} + \frac{iQ_z\left(x + iy\right)}{2}$$

* Harmonics for rank 2
$\vec{\mathbb{Q}}_2^{(1,-1)}[q](A)$
** symmetry

$$-\frac{x^2}{2} - \frac{y^2}{2} + z^2$$

** expression

$$-\frac{\sqrt{6}Q_x x}{6} - \frac{\sqrt{6}Q_y y}{6} + \frac{\sqrt{6}Q_z z}{3}$$

$\vec{\mathbb{Q}}_2^{(1,1)}[q](A)$
** symmetry

$$-\frac{x^2}{2} - \frac{y^2}{2} + z^2$$

** expression

$$-\frac{\sqrt{21}Q_x x\left(x^2 + y^2 - 4z^2\right)}{14} - \frac{\sqrt{21}Q_y y\left(x^2 + y^2 - 4z^2\right)}{14} - \frac{\sqrt{21}Q_z z\left(3x^2 + 3y^2 - 2z^2\right)}{14}$$

$\vec{\mathbb{G}}_2^{(1,0)}[q](A, 1)$
** symmetry

$$\frac{\sqrt{3}x^2}{2} - \frac{\sqrt{3}y^2}{2}$$

** expression

$$\frac{\sqrt{2}Q_x yz}{2} + \frac{\sqrt{2}Q_y xz}{2} - \sqrt{2}Q_z xy$$

$\vec{\mathbb{G}}_2^{(1,0)}[q](A, 2)$
** symmetry

$$\sqrt{3}xy$$

** expression

$$-\frac{\sqrt{2}Q_x xz}{2} + \frac{\sqrt{2}Q_y yz}{2} + \frac{\sqrt{2}Q_z\left(x - y\right)\left(x + y\right)}{2}$$

$\vec{\mathbb{Q}}_2^{(1,-1)}[q](B, 1)$
** symmetry

$$\frac{\sqrt{3}x^2}{2} - \frac{\sqrt{3}y^2}{2}$$

** expression

$$\frac{\sqrt{2}Q_x x}{2} - \frac{\sqrt{2}Q_y y}{2}$$

$\vec{\mathbb{Q}}_2^{(1,-1)}[q](B,2)$
** symmetry

$$\sqrt{3}xy$$

** expression

$$\frac{\sqrt{2}Q_x y}{2} + \frac{\sqrt{2}Q_y x}{2}$$

$\vec{\mathbb{Q}}_2^{(1,1)}[q](B,1)$
** symmetry

$$\frac{\sqrt{3}x^2}{2} - \frac{\sqrt{3}y^2}{2}$$

** expression

$$\frac{\sqrt{7}Q_x x\left(3x^2-7y^2-2z^2\right)}{14} + \frac{\sqrt{7}Q_y y\left(7x^2-3y^2+2z^2\right)}{14} + \frac{5\sqrt{7}Q_z z\left(x-y\right)\left(x+y\right)}{14}$$

$\vec{\mathbb{Q}}_2^{(1,1)}[q](B,2)$
** symmetry

$$\sqrt{3}xy$$

** expression

$$\frac{\sqrt{7}Q_x y\left(4x^2-y^2-z^2\right)}{7} - \frac{\sqrt{7}Q_y x\left(x^2-4y^2+z^2\right)}{7} + \frac{5\sqrt{7}Q_z xyz}{7}$$

$\vec{\mathbb{G}}_2^{(1,0)}[q](B)$
** symmetry

$$-\frac{x^2}{2} - \frac{y^2}{2} + z^2$$

** expression

$$\frac{\sqrt{6}Q_x yz}{2} - \frac{\sqrt{6}Q_y xz}{2}$$

$\vec{\mathbb{Q}}_2^{(1,-1)}[q](E^{(a)})$
** symmetry

$$\frac{\sqrt{6}ixz}{2} + \frac{\sqrt{6}yz}{2}$$

** expression

$$\frac{iQ_x z}{2} + \frac{Q_y z}{2} + \frac{iQ_z\left(x-iy\right)}{2}$$

$\vec{\mathbb{Q}}_2^{(1,1)}[q](E^{(a)})$
** symmetry

$$\frac{\sqrt{6}ixz}{2} + \frac{\sqrt{6}yz}{2}$$

** expression

$$\frac{\sqrt{14}iQ_x z\left(4x^2-5ixy-y^2-z^2\right)}{14} - \frac{\sqrt{14}Q_y z\left(x^2-5ixy-4y^2+z^2\right)}{14} - \frac{\sqrt{14}iQ_z\left(x-iy\right)\left(x^2+y^2-4z^2\right)}{14}$$

$\vec{\mathbb{G}}_2^{(1,0)}[q](E^{(a)})$
** symmetry

$$-\frac{\sqrt{6}ixz}{2} + \frac{\sqrt{6}yz}{2}$$

** expression

$$-\frac{iQ_x\left(xy+iy^2-iz^2\right)}{2} + \frac{iQ_y\left(x^2+ixy-z^2\right)}{2} + \frac{Q_z z\left(x+iy\right)}{2}$$

$\vec{\mathbb{Q}}_2^{(1,-1)}[q](E^{(b)})$
** symmetry

$$-\frac{\sqrt{6}ixz}{2} + \frac{\sqrt{6}yz}{2}$$

** expression

$$-\frac{iQ_xz}{2}+\frac{Q_yz}{2}-\frac{iQ_z\left(x+iy\right)}{2}$$

$\vec{\mathbb{Q}}_2^{(1,1)}[q](E^{(b)})$
** symmetry

$$-\frac{\sqrt{6}ixz}{2}+\frac{\sqrt{6}yz}{2}$$

** expression

$$-\frac{\sqrt{14}iQ_xz\left(4x^2+5ixy-y^2-z^2\right)}{14}-\frac{\sqrt{14}Q_yz\left(x^2+5ixy-4y^2+z^2\right)}{14}+\frac{\sqrt{14}iQ_z\left(x+iy\right)\left(x^2+y^2-4z^2\right)}{14}$$

$\vec{\mathbb{G}}_2^{(1,0)}[q](E^{(b)})$
** symmetry

$$\frac{\sqrt{6}ixz}{2}+\frac{\sqrt{6}yz}{2}$$

** expression

$$\frac{iQ_x\left(xy-iy^2+iz^2\right)}{2}-\frac{iQ_y\left(x^2-ixy-z^2\right)}{2}+\frac{Q_zz\left(x-iy\right)}{2}$$

* Harmonics for rank 3
$\vec{\mathbb{Q}}_3^{(1,-1)}[q](A,1)$
** symmetry

$$\sqrt{15}xyz$$

** expression

$$Q_xyz+Q_yxz+Q_zxy$$

$\vec{\mathbb{Q}}_3^{(1,-1)}[q](A,2)$
** symmetry

$$\frac{\sqrt{15}x^2z}{2}-\frac{\sqrt{15}y^2z}{2}$$

** expression

$$Q_xxz-Q_yyz+\frac{Q_z\left(x-y\right)\left(x+y\right)}{2}$$

$\vec{\mathbb{Q}}_3^{(1,1)}[q](A,1)$
** symmetry

$$\sqrt{15}xyz$$

** expression

$$\frac{\sqrt{15}Q_xyz\left(6x^2-y^2-z^2\right)}{6}-\frac{\sqrt{15}Q_yxz\left(x^2-6y^2+z^2\right)}{6}-\frac{\sqrt{15}Q_zxy\left(x^2+y^2-6z^2\right)}{6}$$

$\vec{\mathbb{Q}}_3^{(1,1)}[q](A,2)$
** symmetry

$$\frac{\sqrt{15}x^2z}{2}-\frac{\sqrt{15}y^2z}{2}$$

** expression

$$\frac{\sqrt{15}Q_xxz\left(5x^2-9y^2-2z^2\right)}{12}+\frac{\sqrt{15}Q_yyz\left(9x^2-5y^2+2z^2\right)}{12}-\frac{\sqrt{15}Q_z\left(x-y\right)\left(x+y\right)\left(x^2+y^2-6z^2\right)}{12}$$

$\vec{\mathbb{G}}_3^{(1,0)}[q](A)$
** symmetry

$$-\frac{3x^2z}{2}-\frac{3y^2z}{2}+z^3$$

** expression

$$-\frac{\sqrt{3}Q_xy\left(x^2+y^2-4z^2\right)}{4}+\frac{\sqrt{3}Q_yx\left(x^2+y^2-4z^2\right)}{4}$$

$\vec{\mathbb{Q}}_3^{(1,-1)}[q](B)$

** symmetry

$$-\frac{3x^2z}{2}-\frac{3y^2z}{2}+z^3$$

** expression

$$-\frac{\sqrt{15}Q_xxz}{5}-\frac{\sqrt{15}Q_yyz}{5}-\frac{\sqrt{15}Q_z\left(x^2+y^2-2z^2\right)}{10}$$

$\vec{\mathbb{Q}}_3^{(1,1)}[q](B)$
** symmetry

$$-\frac{3x^2z}{2}-\frac{3y^2z}{2}+z^3$$

** expression

$$-\frac{5Q_xxz\left(3x^2+3y^2-4z^2\right)}{12}-\frac{5Q_yyz\left(3x^2+3y^2-4z^2\right)}{12}+\frac{Q_z\left(3x^4+6x^2y^2-24x^2z^2+3y^4-24y^2z^2+8z^4\right)}{12}$$

$\vec{\mathbb{G}}_3^{(1,0)}[q](B,1)$
** symmetry

$$\sqrt{15}xyz$$

** expression

$$\frac{\sqrt{5}Q_xx\left(y-z\right)\left(y+z\right)}{2}-\frac{\sqrt{5}Q_yy\left(x-z\right)\left(x+z\right)}{2}+\frac{\sqrt{5}Q_zz\left(x-y\right)\left(x+y\right)}{2}$$

$\vec{\mathbb{G}}_3^{(1,0)}[q](B,2)$
** symmetry

$$\frac{\sqrt{15}x^2z}{2}-\frac{\sqrt{15}y^2z}{2}$$

** expression

$$\frac{\sqrt{5}Q_xy\left(x^2-y^2+2z^2\right)}{4}-\frac{\sqrt{5}Q_yx\left(x^2-y^2-2z^2\right)}{4}-\sqrt{5}Q_zxyz$$

$\vec{\mathbb{Q}}_3^{(1,-1)}[q](E^{(a)},1)$
** symmetry

$$\frac{\sqrt{2}x^3}{2}-\frac{3\sqrt{2}ix^2y}{4}-\frac{3\sqrt{2}xy^2}{4}-\frac{3\sqrt{2}xz^2}{4}+\frac{\sqrt{2}iy^3}{2}-\frac{3\sqrt{2}iyz^2}{4}$$

** expression

$$\frac{\sqrt{30}Q_x\left(2x^2-2ixy-y^2-z^2\right)}{20}-\frac{\sqrt{30}iQ_y\left(x^2-2ixy-2y^2+z^2\right)}{20}-\frac{\sqrt{30}Q_zz\left(x+iy\right)}{10}$$

$\vec{\mathbb{Q}}_3^{(1,-1)}[q](E^{(a)},2)$
** symmetry

$$\frac{\sqrt{30}ix^2y}{4}+\frac{\sqrt{30}xy^2}{4}-\frac{\sqrt{30}xz^2}{4}-\frac{\sqrt{30}iyz^2}{4}$$

** expression

$$\frac{\sqrt{2}iQ_x\left(2xy-iy^2+iz^2\right)}{4}+\frac{\sqrt{2}iQ_y\left(x^2-2ixy-z^2\right)}{4}-\frac{\sqrt{2}Q_zz\left(x+iy\right)}{2}$$

$\vec{\mathbb{Q}}_3^{(1,1)}[q](E^{(a)},1)$
** symmetry

$$\frac{\sqrt{2}x^3}{2}-\frac{3\sqrt{2}ix^2y}{4}-\frac{3\sqrt{2}xy^2}{4}-\frac{3\sqrt{2}xz^2}{4}+\frac{\sqrt{2}iy^3}{2}-\frac{3\sqrt{2}iyz^2}{4}$$

** expression

$$\frac{\sqrt{2}Q_x\left(8x^4-15ix^3y-24x^2y^2-24x^2z^2+20ixy^3-15ixyz^2+3y^4+6y^2z^2+3z^4\right)}{24}$$
$$+\frac{\sqrt{2}iQ_y\left(3x^4-20ix^3y-24x^2y^2+6x^2z^2+15ixy^3+15ixyz^2+8y^4-24y^2z^2+3z^4\right)}{24}+\frac{5\sqrt{2}Q_zz\left(4x^3-3ix^2y-3xy^2-3xz^2+4iy^3-3iyz^2\right)}{24}$$

$\vec{\mathbb{Q}}_3^{(1,1)}[q](E^{(a)},2)$
** symmetry

$$\frac{\sqrt{30}ix^2y}{4}+\frac{\sqrt{30}xy^2}{4}-\frac{\sqrt{30}xz^2}{4}-\frac{\sqrt{30}iyz^2}{4}$$

** expression

$$\frac{\sqrt{30}iQ_x\left(5x^3y-6ix^2y^2+6ix^2z^2-2xy^3-9xyz^2+iy^4-iz^4\right)}{24}$$
$$-\frac{\sqrt{30}iQ_y\left(x^4-2ix^3y-6x^2y^2+5ixy^3-9ixyz^2+6y^2z^2-z^4\right)}{24}+\frac{\sqrt{30}Q_zz\left(2x^3+9ix^2y+9xy^2-5xz^2+2iy^3-5iyz^2\right)}{24}$$

$\vec{\mathbb{G}}_3^{(1,0)}[q](E^{(a)},1)$

** symmetry

$$\frac{\sqrt{2}x^3}{2}+\frac{3\sqrt{2}ix^2y}{4}-\frac{3\sqrt{2}xy^2}{4}-\frac{3\sqrt{2}xz^2}{4}-\frac{\sqrt{2}iy^3}{2}+\frac{3\sqrt{2}iyz^2}{4}$$

** expression

$$-\frac{\sqrt{6}iQ_xz\left(x^2-4y^2+z^2\right)}{8}+\frac{\sqrt{6}Q_yz\left(4x^2-y^2-z^2\right)}{8}+\frac{\sqrt{6}iQ_z\left(x^3+4ix^2y-4xy^2+xz^2-iy^3-iyz^2\right)}{8}$$

$\vec{\mathbb{G}}_3^{(1,0)}[q](E^{(a)},2)$

** symmetry

$$-\frac{\sqrt{30}ix^2y}{4}+\frac{\sqrt{30}xy^2}{4}-\frac{\sqrt{30}xz^2}{4}+\frac{\sqrt{30}iyz^2}{4}$$

** expression

$$\frac{\sqrt{10}iQ_xz\left(x^2+4ixy+2y^2-z^2\right)}{8}+\frac{\sqrt{10}Q_yz\left(2x^2-4ixy+y^2-z^2\right)}{8}-\frac{\sqrt{10}iQ_z\left(x^3+2ix^2y-2xy^2-xz^2-iy^3+iyz^2\right)}{8}$$

$\vec{\mathbb{Q}}_3^{(1,-1)}[q](E^{(b)},1)$

** symmetry

$$\frac{\sqrt{2}x^3}{2}+\frac{3\sqrt{2}ix^2y}{4}-\frac{3\sqrt{2}xy^2}{4}-\frac{3\sqrt{2}xz^2}{4}-\frac{\sqrt{2}iy^3}{2}+\frac{3\sqrt{2}iyz^2}{4}$$

** expression

$$\frac{\sqrt{30}Q_x\left(2x^2+2ixy-y^2-z^2\right)}{20}+\frac{\sqrt{30}iQ_y\left(x^2+2ixy-2y^2+z^2\right)}{20}-\frac{\sqrt{30}Q_zz\left(x-iy\right)}{10}$$

$\vec{\mathbb{Q}}_3^{(1,-1)}[q](E^{(b)},2)$

** symmetry

$$-\frac{\sqrt{30}ix^2y}{4}+\frac{\sqrt{30}xy^2}{4}-\frac{\sqrt{30}xz^2}{4}+\frac{\sqrt{30}iyz^2}{4}$$

** expression

$$-\frac{\sqrt{2}iQ_x\left(2xy+iy^2-iz^2\right)}{4}-\frac{\sqrt{2}iQ_y\left(x^2+2ixy-z^2\right)}{4}-\frac{\sqrt{2}Q_zz\left(x-iy\right)}{2}$$

$\vec{\mathbb{Q}}_3^{(1,1)}[q](E^{(b)},1)$

** symmetry

$$\frac{\sqrt{2}x^3}{2}+\frac{3\sqrt{2}ix^2y}{4}-\frac{3\sqrt{2}xy^2}{4}-\frac{3\sqrt{2}xz^2}{4}-\frac{\sqrt{2}iy^3}{2}+\frac{3\sqrt{2}iyz^2}{4}$$

** expression

$$\frac{\sqrt{2}Q_x\left(8x^4+15ix^3y-24x^2y^2-24x^2z^2-20ixy^3+15ixyz^2+3y^4+6y^2z^2+3z^4\right)}{24}$$
$$-\frac{\sqrt{2}iQ_y\left(3x^4+20ix^3y-24x^2y^2+6x^2z^2-15ixy^3-15ixyz^2+8y^4-24y^2z^2+3z^4\right)}{24}+\frac{5\sqrt{2}Q_zz\left(4x^3+3ix^2y-3xy^2-3xz^2-4iy^3+3iyz^2\right)}{24}$$

$\vec{\mathbb{Q}}_3^{(1,1)}[q](E^{(b)},2)$

** symmetry

$$-\frac{\sqrt{30}ix^2y}{4}+\frac{\sqrt{30}xy^2}{4}-\frac{\sqrt{30}xz^2}{4}+\frac{\sqrt{30}iyz^2}{4}$$

** expression

$$-\frac{\sqrt{30}iQ_x\left(5x^3y+6ix^2y^2-6ix^2z^2-2xy^3-9xyz^2-iy^4+iz^4\right)}{24}$$
$$+\frac{\sqrt{30}iQ_y\left(x^4+2ix^3y-6x^2y^2-5ixy^3+9ixyz^2+6y^2z^2-z^4\right)}{24}+\frac{\sqrt{30}Q_zz\left(2x^3-9ix^2y+9xy^2-5xz^2-2iy^3+5iyz^2\right)}{24}$$

$\vec{\mathbb{G}}_3^{(1,0)}[q](E^{(b)},1)$

** symmetry

$$\frac{\sqrt{2}x^3}{2}-\frac{3\sqrt{2}ix^2y}{4}-\frac{3\sqrt{2}xy^2}{4}-\frac{3\sqrt{2}xz^2}{4}+\frac{\sqrt{2}iy^3}{2}-\frac{3\sqrt{2}iyz^2}{4}$$

** expression

$$\frac{\sqrt{6}iQ_xz\left(x^2-4y^2+z^2\right)}{8}+\frac{\sqrt{6}Q_yz\left(4x^2-y^2-z^2\right)}{8}-\frac{\sqrt{6}iQ_z\left(x^3-4ix^2y-4xy^2+xz^2+iy^3+iyz^2\right)}{8}$$

$\vec{\mathbb{G}}_3^{(1,0)}[q](E^{(b)},2)$

** symmetry

$$\frac{\sqrt{30}ix^2y}{4}+\frac{\sqrt{30}xy^2}{4}-\frac{\sqrt{30}xz^2}{4}-\frac{\sqrt{30}iyz^2}{4}$$

** expression

$$-\frac{\sqrt{10}iQ_xz\left(x^2-4ixy+2y^2-z^2\right)}{8}+\frac{\sqrt{10}Q_yz\left(2x^2+4ixy+y^2-z^2\right)}{8}+\frac{\sqrt{10}iQ_z\left(x^3-2ix^2y-2xy^2-xz^2+iy^3-iyz^2\right)}{8}$$